AQUABOTS - EMI Filtering BOM & Placement Guide

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This document specifies all capacitors, ferrite components, and copper tape shielding required for EMI filtering across the AQUABOTS power and signal systems. Each item includes exact placement location, which side (input/output), and distance from target component.

1. BILL OF MATERIALS

#	Component	Value / Rating	Qty	Location	Status
C1	Electrolytic Cap (radial)	1000uF / 35V+	1	Busbar (across +/- studs)	ORDER
C2	Ceramic Disc Cap (thru-hole)	100nF (0.1uF) / 50V+	10+	Multiple (see Sec. 2)	ON HAND *
C3	Electrolytic Cap (radial)	470uF / 35V+ LOW-ESR	1	ESC power input	ORDER
C4	Electrolytic Cap (radial)	10uF / 10V+	1	ESP32 VIN pins	ORDER
C5	Electrolytic Cap (radial)	100uF / 50V	1	Jetson DC barrel input	ON HAND
C6	Electrolytic Cap (radial)	100uF / 50V	1	BEC output / servo conn.	ON HAND
C7	Electrolytic Cap (radial)	100uF / 50V	1	Starlink DC input	ON HAND
F1	Ferrite Toroid (ring core)	19mm ID, MnZn, 38x19x13mm	1	ESC (+) lead, near ESC	ORDER
F2	Clip-on Ferrite (snap)	13mm ID	2	5V Buck input, 12V Buck input	ORDER
F3	Clip-on Ferrite (snap)	13mm ID	1	Starlink (+) input lead	ORDER
F4	Clip-on Ferrite (snap)	13mm ID (or 7mm)	1	5V Buck output to ESP32	ORDER
F5	Axial Ferrite Bead (thru-hole)	~70 Ohm @ 100MHz, 3.5x6mm	6+	I2C SDA/SCL (4), opt UART (2)	ORDER
SH1	Copper Foil Tape (adhesive)	1" wide, conductive adhesive	1 roll	Cable shielding (see Sec. 3)	ORDER
SH2	Auto Electrical Self-Adhesion Tape	Standard	1 roll	Insulation over copper tape	ON HAND
SH3	Split Loom Wrap	1/4" and 3/8" ID	as needed	Outer mechanical protection	ON HAND

Items to Order:

Item	Suggested Product	Est. $
C1: 1000uF/35V	Any radial electrolytic, 35V+	~$2
C3: 470uF/35V LOW-ESR	Low-ESR specifically — only cap where this matters	~$3
C4: 10uF/10V	Any radial electrolytic, 10V+	~$0.50
F1: Ferrite toroid	Hondark 38x19x13mm MnZn (Amazon, 4-pack $12.99)	~$13
F2+F3+F4: Clip-on ferrites	Raogoodcx 13mm 15-pack (Amazon, $15.99)	~$16
F5: Axial ferrite beads	BOJACK 3.5x6x0.8mm 100-pack (Amazon, ~$7)	~$7
SH1: Copper foil tape	1" wide, conductive adhesive, Amazon	~$8

Order total: ~$50. Includes generous spare counts on ferrites and beads.

On-Hand Substitution Notes:

C5, C6, C7: Using 100uF/50V electrolytics already on hand. Voltage rating is a maximum, not a target — 50V on a 5V, 12V, or 22V rail is perfectly fine. Higher rating = more headroom, never harmful.

C2 (ceramic 100nF): Using caps already on hand. Verify voltage rating is 50V+ for any cap placed on the busbar (22–26V rail). Spikes from the ESC can exceed 30V. Caps on the 5V or 12V side are fine at 25V+.

Low-ESR note: Low-ESR is ONLY critical for C3 (470uF at ESC input) because it handles high-current transients from ESC switching. All other electrolytic positions are fine with standard ESR. Using low-ESR everywhere is fine but not worth paying extra for.

Ferrite toroid (F1): Oversized toroids are fine. Impedance comes from the ferrite material (MnZn), not from a tight fit. The 19mm ID on the Hondark toroid easily fits 8 AWG (~3.3mm OD). If choosing between sizes, always go larger. Only failure mode is too small to fit the wire.

Axial ferrite beads (F5): 600 Ohm @ 100MHz through-hole beads do not exist as a consumer product. Generic axial beads (~70 Ohm @ 100MHz) are adequate for short I2C runs. Multiple beads in series do add impedance roughly linearly, but 2-3 max per line — more creates a fragile solder chain. One bead per line + 100nF cap at ESP32 pin is sufficient.

2. PLACEMENT INSTRUCTIONS (by location)

Every cap and ferrite placement is described below with exact position, orientation, and distance from the target component. 'Parallel caps' means two caps soldered across the same two points (+) and (-), side by side. The electrolytic handles low-frequency transients; the ceramic handles high-frequency noise. They MUST both be present to cover the full spectrum.

2.1 BUSBAR (highest priority - do this first)

Part	Side	Exact Placement	How to Install
C1: 1000uF/35V+ Electrolytic	N/A (across rails)	Solder ring terminals to cap leads. Mount on (+) and (-) busbar studs directly. Within 1" of stud centers.	Observe polarity: long lead / stripe = (+). Twist or keep leads as short as possible (<1"). Secure with zip tie to prevent vibration.
C2: 100nF/50V+ Ceramic (ON HAND)	N/A (across rails)	Solder directly in PARALLEL with C1, on the same busbar studs. Leads within 0.5" of stud centers.	No polarity. Solder to same ring terminals as C1 or use separate small ring terminals. Keep leads as short as physically possible.

Why parallel: The 1000uF electrolytic absorbs low-frequency surges (motor startup, ESC switching bursts). The 100nF ceramic absorbs high-frequency hash (>1MHz) that the electrolytic is too slow to filter. Without both, you only filter half the spectrum.

2.2 ESC / MOTOR BRANCH (P5/P6 - 8 AWG)

Part	Side	Exact Placement	How to Install
F1: Ferrite Toroid 19mm ID MnZn (Hondark 38x19x13)	INPUT side (busbar end)	On the 8 AWG RED (+) wire, between MIDI 60A fuse output and ESC power input. Within 2" of ESC input terminal.	Pass the 8 AWG wire THROUGH the toroid center hole once (single pass). Do NOT wrap multiple turns - one pass is enough at these currents. Secure with zip tie.
C3: 470uF/35V+ LOW-ESR Electrolytic (ONLY low-ESR cap)	INPUT side (at ESC terminals)	Across ESC (+) and (-) power input terminals. Solder directly to ESC input pads or to the wires within 0.5" of the ESC connector.	Observe polarity. Low-ESR is critical HERE ONLY — ESC dumps high-current transients that standard caps are too slow to absorb.
C2: 100nF/50V+ Ceramic (ON HAND)	INPUT side (at ESC terminals)	In PARALLEL with C3, across ESC (+) and (-) input. Same solder points as C3.	No polarity. Solder leads directly to C3 leads or to same wire junction. Keep ceramic cap leads < 0.5".

2.3 5V BUCK / ESP32 BRANCH (P13/P14 input, P21 output)

This is your most EMI-sensitive branch. The ESP32's 2.4GHz RF front end and the analog sensor inputs are both vulnerable to conducted noise from the buck converter.

Part	Side	Exact Placement	How to Install
F2: Clip-on Ferrite 13mm (Raogoodcx)	INPUT side (busbar to buck)	On the 12 AWG RED (+) wire (P13) between ATO 1A fuse output and 5V buck converter input. Within 2" of buck input terminal.	Snap the clip-on ferrite around the RED (+) wire only (not both wires). Close and secure with tape or zip tie.
F4: Clip-on Ferrite 13mm (or 7mm)	OUTPUT side (buck to ESP32)	On the 22 AWG pair (P21) between 5V buck output and ESP32 VIN. Within 2" of the ESP32 board.	Pass BOTH the (+) and (-) output wires through the ferrite together. This creates common-mode filtering.
C4: 10uF/10V+ Electrolytic	OUTPUT side (at ESP32 VIN)	Across ESP32 VIN (+) and GND pins. Solder within 0.5" of the ESP32 power input pins/header.	Observe polarity. If using USB-C power input, solder across the 5V and GND pins on the dev board header.

Part	Side	Exact Placement	How to Install
C2: 100nF/50V+ Ceramic (ON HAND)	OUTPUT side (at ESP32 VIN)	In PARALLEL with C4, across ESP32 VIN and GND. Same solder points. Within 0.5" of ESP32 pins.	No polarity. Solder as close to the ESP32 board as physically possible. Shorter leads = better HF filtering.

2.4 12V BUCK / JETSON BRANCH (P11/P12 input, P19 output)

Part	Side	Exact Placement	How to Install
F2: Clip-on Ferrite 13mm (Raogoodcx)	INPUT side (busbar to buck)	On the 12 AWG RED (+) wire (P11) between ATO 3A fuse output and 12V buck converter input. Within 2" of buck input terminal.	Snap the clip-on ferrite around the RED (+) wire only. Same type as the 5V buck input ferrite (F2).
C5: 100uF/50V Electrolytic (ON HAND)	OUTPUT side (at Jetson DC input)	Across Jetson DC barrel (+) and (-) terminals. Solder to barrel jack wires within 1" of the plug.	Observe polarity. Solder across the two wires going into the barrel plug, or across the barrel jack breakout.
C2: 100nF/50V+ Ceramic (ON HAND)	OUTPUT side (at Jetson DC input)	In PARALLEL with C5. Same solder joints, within 1" of Jetson plug.	No polarity. Short leads. Solder directly to C5 leads or same junction.

2.5 BEC / SERVO BRANCH (P15/P16 input, P22 output)

Part	Side	Exact Placement	How to Install
C6: 100uF/50V Electrolytic (ON HAND)	OUTPUT side (at servo connector)	Across BEC output (+) and (-) wires, within 1" of the servo 3-pin connector. NOT at the BEC board itself.	Observe polarity. Solder across the power and ground wires of the servo cable. Shrink-tube the leads.
C2: 100nF/50V+ Ceramic (ON HAND)	OUTPUT side (at servo connector)	In PARALLEL with C6. Same solder junction within 1" of servo plug.	No polarity. Short leads.

2.6 STARLINK BRANCH (P7/P8 - direct from busbar)

Part	Side	Exact Placement	How to Install
F3: Clip-on Ferrite 13mm (Raogoodcx)	INPUT side (busbar to Starlink)	On the 12 AWG RED (+) wire (P7) between ATO 7.5A fuse output and Starlink DC input connector. Within 3" of the fuse block.	Snap around the RED (+) wire only. Starlink has its own internal power conditioning but this prevents ESC noise from coupling in.
C7: 100uF/50V Electrolytic (ON HAND)	OUTPUT side (at Starlink DC wires)	Across Starlink DC input (+) and (-) wires, within 1" of the Starlink power connector.	Observe polarity. Solder across the pigtail wires going to Starlink. Shrink-tube all exposed joints.
C2: 100nF/50V+ Ceramic (ON HAND)	OUTPUT side (at Starlink DC plug)	In PARALLEL with C7. Same solder junction near Starlink plug.	No polarity. Short leads.

2.7 SENSOR SIGNAL LINES (S7-S12 - 26 AWG)

Part	Side	Exact Placement	How to Install
C2: 100nF/50V+ Ceramic (ON HAND)	At ESP32 pin (receiving end)	S10 (TDS Analog): Between ESP32 analog input pin and GND. Within 0.25" of the ESP32 pin. THIS IS THE MOST CRITICAL sensor cap.	Solder directly to ESP32 header pins or to wires at the DuPont connector. Shortest possible leads. This prevents ESC noise from corrupting ADC readings.
F5: Axial Ferrite Bead ~70 Ohm (BOJACK)	In SERIES on SDA and SCL	S7 (IMU I2C) and S8 (pH I2C): One bead in series on each SDA line, one bead in series on each SCL line. 4 beads total. At ESP32 end, within 1".	Cut the 26 AWG signal wire, solder the bead inline (wire-bead-wire). Shrink-tube each joint. Beads are non-polarized. Looks like a tiny resistor.
C2: 100nF/50V+ Ceramic (ON HAND)	At ESP32 pin (receiving end)	S9 (1-Wire DS18B20) and S11 (UART GPS): Between signal pin and GND at ESP32. Within 0.5" of ESP32 pin.	Lower priority than TDS (S10) and I2C (S7/S8). Install if you have spare 100nF caps.

3. COPPER TAPE CABLE SHIELDING

Copper foil tape wrapped around cable bundles provides shielding against radiated EMI. Each shielded bundle is wrapped independently (not combined with other bundles), then layered: copper tape innermost, automotive self-adhesion tape over it for insulation, split loom wrap outermost for mechanical protection.

CRITICAL: The copper shield MUST be grounded at ONE end only, or it acts as an antenna and makes EMI worse. Ground at the SOURCE end (busbar/converter side). Leave the LOAD end ungrounded. Grounding both ends creates a ground loop that injects noise.

Cable	Priority	Ground Drain To	Notes
P21: 5V Buck output to ESP32 (22 AWG pair)	HIGH (do this)	Solder short pigtail from copper tape to 5V buck GND output terminal.	Most EMI-sensitive load. Passes through zone near ESC/motor wires. Shield the twisted pair, ground at buck converter end.
S10: TDS Analog (26 AWG, 12" run)	HIGH (do this)	Solder pigtail from copper tape to ESP32 GND pin or nearby ground point.	Longest analog sensor run. Most vulnerable to radiated pickup from motor. Ground at ESP32 end (source of the ground reference).
S7-S9, S11: Other sensor wires (26 AWG)	MEDIUM (recommended)	Solder pigtail from copper tape to ESP32 GND.	Can bundle all sensor wires together in one copper-wrapped group since they all go to the same ESP32 board. Single drain.
S1: UART to Pixhawk (22 AWG, JST-SH)	MEDIUM (recommended)	Solder pigtail from copper tape to ESP32 GND.	MAVLink data. Shield if cable routes near power wires.
S2: UART to Jetson (22 AWG, JST-SH)	MEDIUM (recommended)	Solder pigtail from copper tape to ESP32 GND.	UART 115200 data. Shield if cable routes near power wires.
P5/P6: ESC to Motor (8 AWG 3-phase)	SKIP	—	These are the noise SOURCE, not victim. Ferrite toroid + caps handle this branch. Shielding 54A wires is impractical.
P1-P4: 4 AWG trunk	SKIP	—	Short, fat, low-impedance. Not worth shielding. Distance from signal wires is the better mitigation.

Wrapping Procedure (per cable bundle):

1. Twist the pair (+ and - wires, or signal + ground) before wrapping. ~2 twists per inch.

2. Wrap copper foil tape around the twisted pair in a spiral with ~50% overlap. Conductive adhesive side facing in. Leave 0.5" of bare wire exposed at each connector end to avoid shorting the copper tape to pins.

3. Solder ground drain pigtail: At the SOURCE end only, solder a short piece of wire (22-26 AWG, 1-2") from the copper tape surface to the local ground point. One connection per shield.

4. Wrap automotive self-adhesion tape over the copper tape for electrical insulation. This prevents the copper shield from contacting other conductors or the aluminum plate.

5. Slide split loom wrap over the whole assembly for abrasion/vibration protection.

6. Secure with zip ties at both ends and every 4-6" along the run.

4. SOLDERING CHEAT SHEET

Use this table as a quick reference while soldering. Check off each item as you install it.

Done	Location	What to Solder/Install	Parallel?	Distance Rule
[ ]	Busbar studs	1000uF/35V+ electrolytic + 100nF ceramic	YES (same studs)	< 1" from stud centers
[ ]	ESC (+) lead	F1 ferrite toroid on 8 AWG RED (single pass)	N/A (in series)	< 2" from ESC input
[ ]	ESC input terminals	470uF/35V+ LOW-ESR electrolytic + 100nF ceramic	YES (same terminals)	< 0.5" from ESC pads
[ ]	5V Buck input (+)	F2 clip-on ferrite on 12 AWG RED	N/A (in series)	< 2" from buck input
[ ]	5V Buck output	F4 clip-on ferrite on 22 AWG pair (both wires)	N/A (common-mode)	< 2" from ESP32 board
[ ]	ESP32 VIN pins	10uF/10V+ electrolytic + 100nF ceramic	YES (same pins)	< 0.5" from ESP32 pins
[ ]	12V Buck input (+)	F2 clip-on ferrite on 12 AWG RED	N/A (in series)	< 2" from buck input
[ ]	Jetson DC barrel	100uF/50V electrolytic (ON HAND) + 100nF ceramic	YES (same wires)	< 1" from barrel plug
[ ]	Servo connector	100uF/50V electrolytic (ON HAND) + 100nF ceramic	YES (same wires)	< 1" from servo plug
[ ]	Starlink (+) lead	F3 clip-on ferrite on 12 AWG RED	N/A (in series)	< 3" from ATO fuse block
[ ]	Starlink DC input	100uF/50V electrolytic (ON HAND) + 100nF ceramic	YES (same wires)	< 1" from Starlink plug
[ ]	TDS analog pin S10	100nF ceramic pin-to-GND	No (single cap)	< 0.25" from ESP32 pin
[ ]	I2C SDA/SCL S7,S8	F5 axial ferrite bead inline (x4 total)	N/A (in series)	< 1" from ESP32 pins
[ ]	1-Wire/UART S9,S11	100nF ceramic pin-to-GND (optional)	No (single cap)	< 0.5" from ESP32 pin
[ ]	P21 cable (ESP32)	Copper tape shield + GND drain at buck end	N/A (shielding)	Full cable run
[ ]	S10 cable (TDS)	Copper tape shield + GND drain at ESP32 end	N/A (shielding)	Full cable run
[ ]	Sensor bundle	Copper tape shield + GND drain at ESP32 end	N/A (shielding)	Full cable run
[ ]	S1, S2 UARTs	Copper tape shield + GND drain at ESP32 end	N/A (shielding)	Full cable run (optional)

5. GENERAL RULES

1. **Lead length matters.** Every millimeter of wire between a cap and the pin it protects is an inductor that degrades filtering. Cut leads as short as possible. The ceramic cap is useless at HF if its leads are 2 inches long.

2. **Electrolytic polarity.** The stripe / shorter lead is always (-). Reversing polarity on an electrolytic will cause it to fail, possibly violently. Double-check every one.

3. **Ceramic caps have no polarity.** Solder them either way.

4. **Voltage rating is a maximum, not a target.** A 50V cap on a 5V rail is fine. Higher rating = more headroom. Only requirement: rating must exceed the rail voltage plus expected spikes. Use 50V+ on anything touching the 22-26V busbar.

5. **Ferrite clip-ons go on ONE wire** (the positive lead) for differential-mode filtering. Exception: the 5V buck output ferrite (F4) goes around BOTH wires together for common-mode filtering.

6. **Ferrite beads (inline, for sensors)** go in SERIES — you cut the signal wire and solder the bead in between. They are not polarized. They look like tiny resistors with axial leads.

7. **Copper tape shields MUST be grounded at ONE end only.** Ungrounded shield = antenna (makes EMI worse). Both ends grounded = ground loop (injects noise). Source-end ground only.

8. **Parallel caps:** Whenever this document says 'parallel', it means the electrolytic and ceramic are soldered across the exact same (+) and (-) points, side by side. They share the same solder joints.

9. **Install order:** Busbar caps first, then ferrites on branch leads, then output-side caps, then sensor filtering, then copper tape shielding last. Test with multimeter at each load point after installing caps to verify no shorts before energizing.

10. **Vibration:** This is a marine drone. Secure all caps, ferrites, and shield drains with zip ties, shrink tube, or hot glue after soldering. Unsecured components WILL break free.